

Layla Mohamed Khalil

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EDUCATION

Misr International University

Bachelor's Degree in Computer Science (Minor in Artificial Intelligence)

Cairo, Egypt

Sep. 2022 – Jun. 2026

EXPERIENCE

Deloitte

Computer Science Intern

Cairo, Egypt

Jul. 2025 – Sep. 2025

- Completed hands-on training across AI, cloud infrastructure, full-stack development, cybersecurity, and software testing, gaining exposure to industry practices and technologies.
- Gained practical exposure to advanced AI concepts and industry practices beyond academic coursework through hands-on learning experiences.
- Contributed to the development of an AI-powered CV screening chatbot using NLP workflows, cloud infrastructure, security, and testing practices.
- Worked with cloud and on-premises environments, gaining experience in deployment, scalability, and system management.

NorthKraft UG

Frontend Developer

Remote, Clausthal-Zellerfeld, Germany

Jan. 2025 – Mar. 2025

- Developed responsive React.js and Next.js interfaces while collaborating remotely with designers and backend engineers to deliver scalable web applications.

matchcare GmbH

Full-Stack Developer

Remote, Berlin, Germany

Sep. 2024 – Nov. 2024

- Developed and optimized React/Next.js features for a production healthcare platform, improving responsiveness, UX, and maintainability.
- Collaborated remotely with cross-functional teams to implement scalable functionality and improve application performance through component optimization.

Misr International University

Co-Assistant

Cairo, Egypt

Oct. 2024 – Jan. 2025

- Assisted programming labs and mentored students in debugging, algorithms, problem-solving, and software development concepts.

PROJECTS

ReVerie – AI-Powered VR Therapeutic Platform for Alzheimer's Patients | Python, CV, Deep Learning, GCP, Supabase, Unreal Engine

- Co-developing an AI-powered VR therapeutic platform within a multidisciplinary team to transform personal photographs into immersive reminiscence experiences for Alzheimer's patients.
- Contributed to memory reconstruction workflows using depth estimation, segmentation, inpainting, 3D reconstruction, and scene assembly for immersive VR experiences.
- Developed scalable GCP-based AI pipelines integrated with Supabase and contributed to Flutter mobile/web interfaces and Unreal Engine visualization for supervised patient-doctor interactions.

HR Chatbot – Talent Acquisition Assistant (Deloitte Internship Project) | LangChain, Next.js, GCP, RAG

- Collaborated with a multidisciplinary team during my Deloitte internship to develop an AI-powered HR chatbot using LangChain, RAG, Next.js, and GCP to support CV screening and recruitment Q&A.

AI-Powered Smart Waste Segregation System | Deep Learning, Computer Vision, Arduino

- Developed a MobileNetV3 waste classification system integrated with ESP32-CAM, Arduino, and automated sorting hardware.
- Fine-tuned DenseNet121 on a 2,527-image, 4-class garbage dataset, achieving 98% test accuracy and 97.89% macro F1-score.
- Implemented A* pathfinding for optimized waste collection and routing operations.

Credit Card Approval Prediction | Machine Learning, Genetic Algorithms, GridSearchCV

- Developed an ML pipeline using preprocessing, feature engineering, statistical analysis, and Genetic Algorithm-based feature selection.
- Compared Decision Tree, KNN, and MLP models, achieving 91% accuracy through hyperparameter optimization.

Emotion Detection Using NLP | Machine Learning, Deep Learning, NLP

- Applied 6 machine learning models and 4 deep learning models to classify emotions from text.

Optimizing Network Flow Using Edmonds-Karp Algorithm | Greedy Algorithms, Dynamic Programming

- Implemented and evaluated three Edmonds-Karp variations for maximum flow optimization and performance trade-off analysis.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, PHP, Dart, SQL

Web & Mobile: React.js, Next.js, Node.js, Flutter, HTML, CSS

AI/ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI, Retrieval-Augmented Generation (RAG), TensorFlow, PyTorch, Scikit-learn, LangChain, Data Preprocessing, Feature Engineering

Cloud & Tools: Google Cloud Platform (GCP), AWS, Git, MongoDB, Supabase, Jupyter Notebook

Data Science: NumPy, Pandas, SciPy, Data Analysis, Statistical Analysis

Algorithms & CS Fundamentals: Data Structures, Algorithms, Optimization Techniques, Distributed Systems, Software Engineering Principles, Network Security

RESEARCH

Emotion Detection Using NLP (IEEE Published Research)

- Co-authored IEEE-published research on text-based emotion detection using machine learning and deep learning models.

AI-Driven Digital Therapeutic Interventions for Early-Stage Alzheimer's Disease

- Submitted to *Complexity* (Wiley), 2025; systematic review of AI-driven memory reconstruction, VR reminiscence therapy, computer vision, digital twins, and cognitive stimulation technologies for Alzheimer's disease.

ReVerie: Personalized AI-Based 2D-to-3D Memory Reconstruction for Immersive Reminiscence Therapy

- Submitted to MIU Computer Conference (MIUCC'26); proposed a personalized AI pipeline that transforms patient photographs into immersive VR environments using depth estimation, segmentation, inpainting, and 3D reconstruction.